

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • Stamp 1G — Full-Time Work Eligible

PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Defined and developed **DFT (Design for Testability)** strategies to enhance product testability and reduce manufacturing costs for mixed-signal integrated circuits.
- Executed test development phases, including silicon functionality verification, **characterization**, and yield analysis on high-performance **ATE (Automated Test Equipment)** platforms.
- Supported the rapid ramp-up of ultra-high-volume new products by optimizing test solutions for mass production environments.
- Designed and implemented innovative, reliable, and efficient multi-site **probe and test solutions** for mixed-signal ATE platforms.
- Performed pre-tapeout verification of test solutions using **design simulation tools**, ensuring readiness for silicon validation.
- Collaborated with peers and the wider Test Development community to advance measurement methodologies through continuous improvement initiatives.
- Containerized test application components using **Docker** for consistent deployment across development and production environments on **AWS EC2**.

SKILLS

Hardware Engineering – Analogue Electronics, Digital Electronics, DFT, ATE, Probe and Test Solutions

Software & Scripting – Python, Java, C, Visual Basic

Development Lifecycle – Test Development, Silicon Functionality, Characterization, Yield Analysis, Debug, Production Release

Tools & Platforms – Design Simulation Tools, Jenkins, Git, Jira, Confluence, AWS EC2, Docker

Methodologies – Agile, Scrum, Measurement Methodologies, Continuous Improvement

PROJECTS

Mixed-Signal Test Solution for Capacitive Sensors

- Developed automated test sequences in **Python** for a new capacitive sensing IC, covering functional, performance, and environmental parameters.
- Integrated test results into a CI/CD pipeline using **Jenkins**, enabling automated regression testing and faster feedback loops.
- Designed and validated test patterns that improved test coverage by **15%**, directly contributing to higher product quality.

ATE Platform Optimization for High-Volume Production

- Refactored legacy **ATE test programs** to improve execution speed by an average of **20%**, increasing throughput for mass production.
- Implemented data analysis scripts in **Python** to identify yield detractors and propose targeted improvements for test stations.
- Collaborated with offshore production teams to deploy optimized test solutions, ensuring seamless integration and minimal downtime.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)